8 This question is about the movement of water from the roots of a tree to its leaves.

Water moves up a tree through its vast network of conduits. These conduits are similar to capillary tubes. It is suspected that water moves up the conduits due to low pressure in the conduits which "sucks" the water upwards, or by capillary action, or a combination of both. Capillary action is a phenomenon whereby water rises up a small tube due to upward forces caused by the adhesion of water to the walls of the tube.

To investigate capillary action, a capillary tube, open at both ends, is supported vertically with one end immersed in water, as shown in Fig. 8.1. The water in the narrow bore of the tube forms a column of height *h*.

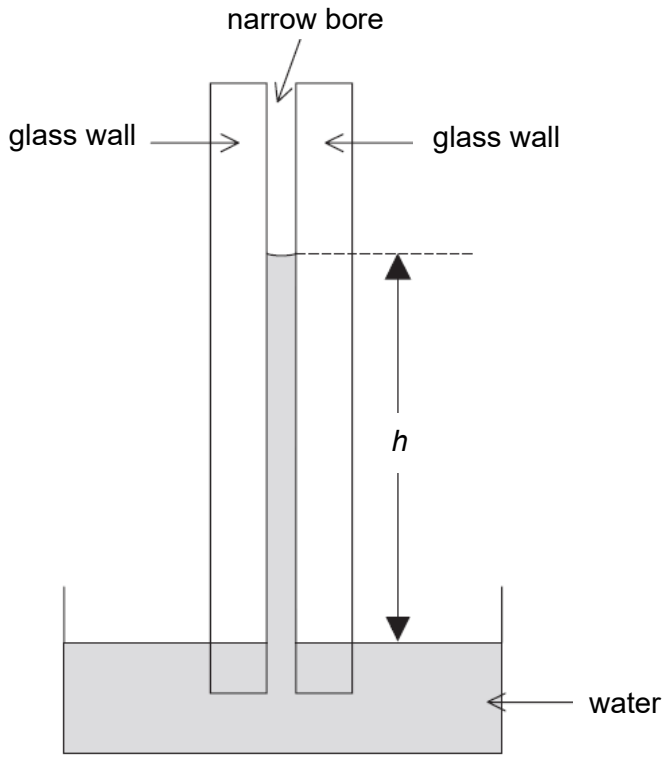

Fig. 8.1
(not to scale)

(a) The height *h* of the water column for a particular capillary tube was measured as the temperature of water θ was varied. Fig. 8.2 shows the data collected.

θ / °C	*h* / cm
30	14.0
40	13.2
50	12.5
60	11.5
70	10.9
80	10.0

Fig. 8.2

Fig. 8.3 shows the variation with temperature θ of height h.

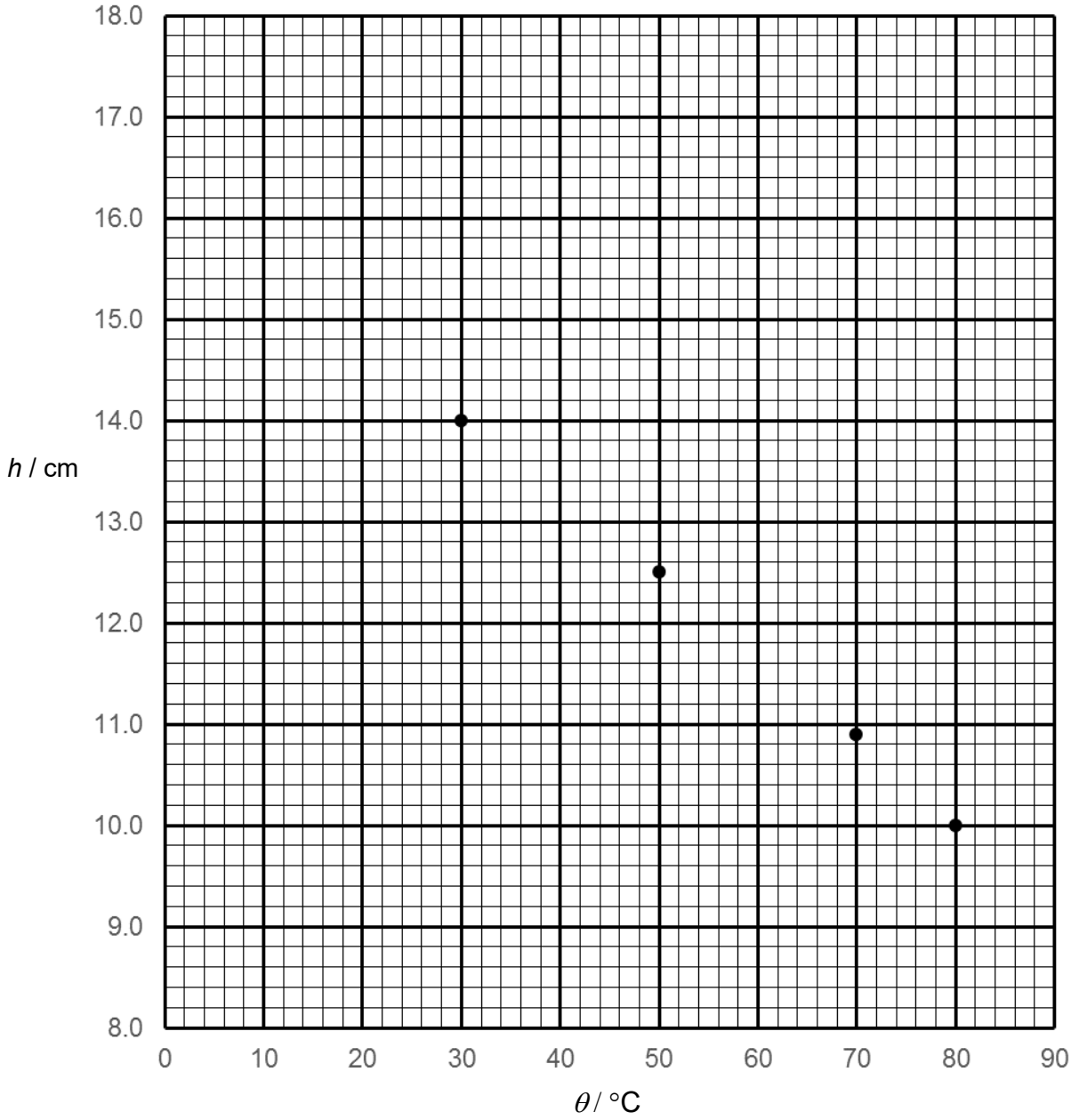

Fig. 8.3

(i) On Fig. 8.3, plot the points for $\theta = 40$ °C and $\theta = 60$ °C. Draw a line of best fit through the data points. [2]

(ii) Using Fig 8.3, determine the height h_0 of the water column when the temperature is 0 °C.

h_0 = ...cm [1]

(iii) It is suggested that the relationship between θ and h is

$$\frac{h}{h_0} = 1 - k\theta$$

where k is a constant.

Explain why the results of this experiment supports the relationship suggested.

.....

.....

.....

..... [3]

(iv) Using the line drawn in **(a)(i)**, determine the value of k, including its units.

k = ... [3]

(b) The experiment is repeated using capillary tubes with bores of different radii r but keeping the water temperature constant. Fig. 8.4 shows the variation with $\frac{1}{r}$ of height h for a water temperature of 20 °C.

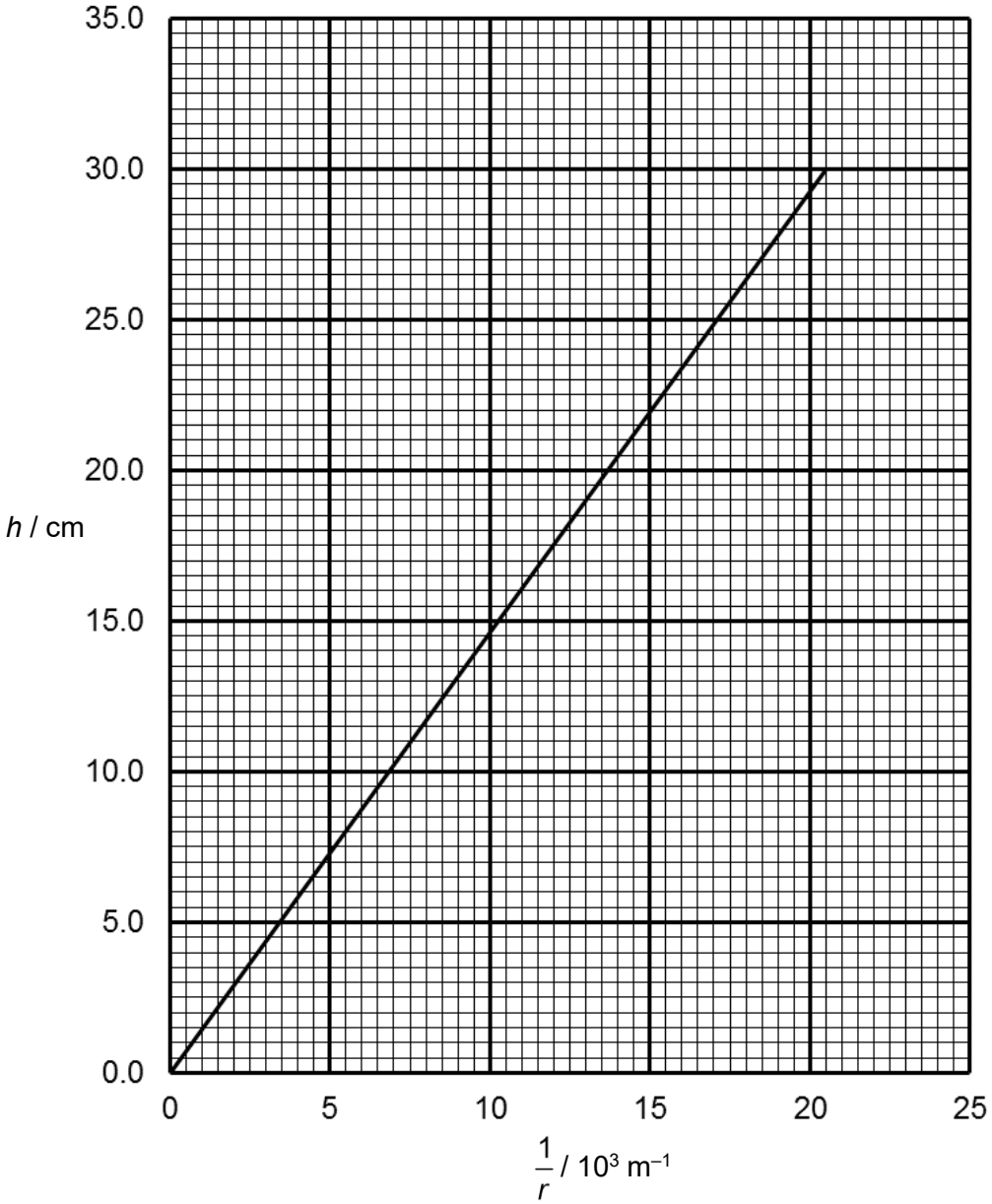

Fig. 8.4

(i) Use Fig. 8.4 to estimate the radius of the bore of the tubes in a 25-metre tall tree, which will enable water to be raised by capillary action from ground level to the top of the tree.

radius = ... m [3]

(ii) State one assumption made in your estimation in **(b)(i)**.

.....

..... [1]

(iii) Comment on your answer obtained in **(b)(i)**.

.....

.....

..... [2]

(c) The other means of moving water up a tree is to create a low pressure in the bore of the tubes in the tree.

(i) Suggest how low pressure can be created in the bore of the tubes in a tree.

.....

..... [1]

(ii) Using the following data, calculate the height which water can be moved up a tree via low pressure in the bore of the tubes.

Atmospheric pressure = 101 kPa

Pressure in the bore of the tubes in the tree = 7.8 kPa

Density of water = 1000 kg m^{-3}

height = ... [2]

(iii) Suggest and explain how the height in **(c)(ii)** will change during a hot day.

.....

.....

..... [2]